MILITARY SPECIFICATION

LAUNCHER, TARGET, LAu-24/A
FOR USE WITH AQM-37A MISSILE TARGET

This specification has been approved by the
NavalAir SystEMs Command, Department of the Navy

1 SCOPE

1.1 Scope. This specification covers the design, production, inspection, and. preparation for delivery of the LAU-24/A Target Launcher.

2 APPLICABLE DocUMENTS

2.1 Government-Furnished Documents, The following documents of the issue in effect on the date of invitation for bid or request for 'proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical
PPP-B-601	Boxes, Wood, Cleated. - Plywood
PPP-B-636	Box, Fiberboard

MILITARY

miL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and. Associated Electrical Equipment
MIL-V-173.	Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment
MIL-C-5015	Connectors, Electric, AN Type
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification
MIL-R-5757	Relays, Electrical (Excluding Thermal) for Electronic and, Communication Type Equipment, General Specification for

FSC 1720

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SPECIFICATIONS , MILITARY (Cent'd)

MIL-H-6088	Heat Treatment of Aluminum Alloys
MIL-R-6106	Relays, Electric, Aircraft, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice, Process for)
MIL-F-7179	Finishes and Coatings, General Specification for Protection of Aircraft and Aircraft Parts
MIL-T-7743	Testing, Store Suspension Equipment, General Specification for
MIL-R-8573	Reservoirs, Air, Nonshatterable Steel
MIL-w-8777	Wire, Electrical, Silicone Insulated, Copper, 600-Volt
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-23931	Cartridge, Impulse, Mark 9Mod O
MIL-P-24014	Preclusion of Hazards from Electromagnetic Radiation to Ordnance, General Requirements for
MIL-c-26482	Connectors, Electrical, Circular, Miniature, Quick-Disconnect

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-sTD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking ofU. S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-838	Lubrication of Military Equipment

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND

NAVAIR DWG. 67A75H1	LAU-24/A Target Launcher for AQM-37A Missile Target; Interface Control Drawing

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

WR-5	Support Equipment Design, Approval, Selection and. Ordering for Bureau of Naval Weapons Contracts
WR-11	Design and Test of Packaging, Packing, Shipping and Handling Equipment for Weapon System (components
WR-38	Mark and Mod Nomenclature System and Assignments to Ancillary Items inAEC Nomenclature System
NAVWEPS 17-15-26	Guided Missile Launcher Electrical Circuit Test Set TTU-183/F

BULLETINS

AIR FORCE - NAVY AERONAUTICAL

ANA 445	Engineering Changes to Weapons, Systems, Equipments and Facilities

(Copies of specifications, standards, and ordnance data requiredby suppliers in connection with specific procurement functions should. be obtained from the Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Copies of WR documents should be obtained from the Naval Ordnance Station, CTDO, Louisville, Kentucky 40214, Copies of drawings should be obtained from the procuring activity or as directed, by the contract.)

2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. The issue in effect on the date of invitation for bid or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H-28	Screw Threads for Federal Services

(Application for copies shouldbe addressedto the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3 REQUIREMENTS

3.1 Description. The LAU-24/A Target Launcher (launcher) depicted in

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3.1 (Cont'd.)

Figure 1 shall provide a reliable method, to mount, carry, and safely air-launch an AQM-37A Missile Target (target) from an aircraft. The launcher shall consist of an aerodynamically contoured structural frame which shall house and provide access to a cartridge-powered trapeze ejector mechanism, retaining hooks, sway braces, an electrical control box, and a system for controlled pressurization of a mounted target, The launcher shall include provisions for safety devices and for physical, electrical, and mechanical mating of the launcher to the supporting aircraft and to the supported target.

3.2 General Requirements.

3.2.1 Materials. All materials entering into the final assembly of the launcher shall conform to the applicable Federal or Military specification. Materials which are not coveredby specification or which are not specifically designated herein shall be of the best quality, of the lightest practicable weight consistent with strength and durability, and entirely suited to the purpose for which intended under all conditions of test and service operation.

3.2.1.1 Standard Parts, Parts such as bolts, nuts, washers, rivets, connectors, and the like shall conform to AN Standards or shall be interchangeable functionally and dimensionally with, or replaceable by AN or MS parts without modification.

3.2.1.2 Dissimilar Metals. The use of dissimilar metals shall be governed. by MIL-STD-454.

3.2.1.3 Flammable Materials. Materials which will support combustion or which are capable of causing an explosion shall not be used.

3.2.1.4 Toxic Materials. Materials which may produce toxic or corrosive effects shall not be used.

3.2.1.5 Magnesium. The use of magnesium shall require the specific approval of the NAVAL AIR SYSTEMS COMMAND (NASC) or its appointed agent.

3.2.2 Fungus Resistance. Except where parts or materials are within hermetically sealed cases, the launcher shall be fungus-proofed by selection of parts and materials that are non-nutrient for fungi. (See Requirement 4, MIL-STD-454.) Parts and materials that are not inert to fungus growth shall be treated prior to use so that overall spraying of the launcher will not be necessary. Such parts shall be considered Type II as described in MIL-T-152 and treated accordingly, using varnish complying with MIL-V-173.

3.2.3 protective Coatings and. Surface Treatments. Protective coatings and surface treatments of the launcher shall be in accordance with the" applicable provisions of MIL-F-7179.

3.2.3.1 Exterior Color. The exterior color of the launcher shall be white, color number 27875 of FED-STD-595.

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3.2. 4 Marking. Parts and assemblies shall be marked in accordance with MIL-STD-130.

3.2.4.1 Nomenclature Assignments and Nameplates. Requests for nomenclature assignments shall be made in accordance with WR-38. Nameplates shall *conform to* MIL-N-18307.

3.2.4.2 Additional Information. Each launcher shall bear the date of manufacture and a serial number which will be assigned by the Government inspector upon request, or as specified in the contract or purchase order.

3.2.5 Lubrication. Provisions shall be made for adequate lubrication of all bearings and wearing surfaces, with due consideration to the operating environments of the launcher as specified herein. The design shall. be such that the launcher will operate satisfactorily with a minimum of re-lubrication and that a minimum of grades and types of lubricants is required. MIL-STD-838 shall be used when selecting lubricants and related materials. Prior to production of the launcher, the contractor shall submit to NASC for approval a list of lubricants selected for use and the application for which each is intended.

3.2.6 Threads. All. threads shall be in accordance with Handbook H-28. The NOT-GO gaging practice shall be as set forth therein.

3.2.7 Electrical Connectors. Electrical connectors shall conform toMIL-c-5015, MIL-C-26482, and MIL-P-24014 as applicable. Each connector shall have a ground-return terminal.

3.2.8 Relays. Relays shall conform to MIL-R-5757 *or* MIL-R-6106, andMIL-P-24014.

3.2.9 Wiring. Installation of wiring in the launcher shall be in accordance with MIL-W-5088 and MIL-P-24014.

3.2.9.1 Wire. Wire used in the launcher shall conform to MIL-W-8777.

3.2.10 Dielectric Strength. When tested in acccordance with 4.4.15, the electrical insulation between each electrical circuit and all other electrical circuits, between each electrical circuit and the structural ground, and between relay contacts shall experience no arcing, increase or fluctuation in leakage current, breaks, *or* other damage.

3.2.11 Insulation Resistance. When tested in accordance with 4.4.16, the insulation resistance of the electrical circuits and devices shall be not less than one megohm.

3.2.12 Heat Treatment. Heat treatment of aluminum alloys shall be in accordance with MIL-H-6088. Steel parts shall be heat-treated in accordance with MIL-H-6875.

3.2.13 Interchangeability. Launcher components shall be interchangeable and shall assemble with mating parts without selection or fitting. (See 4.4.17.)

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3.2.14 Reliability. The launcher shall be capable of launching 500 targets with an operational reliability of 95% per each 100 launches. Servicing and replacement of parts shall be permitted. (See 3.5.12.)

3.2.15 Special Tools. The launcher shall be capable of disassembly and replacement of parts without the use of special tools, except for one multi-purpose tool, which shall. be providedby the contractor in accordance with WR-5, to facilitate the rapid removal of a spent cartridge, installation of replacement cartridge, and disassembly of the combustion chamber and mechanism for cleaning.

3.3 Design and Construction.

3.3.1 General. The launcher shall be designed and constructed within the limitations and other parameters delineated on Naval Air Systems Command Drawing (NAVAIR Dwg.) 67A75H1.

3.3.2 " Weight." The weight. of the launcher shallbe not greater than 195 . pounds. (4.4.19)

3*3*3 Access Doors. The. launcher doors, provided to permit. -reasonable access to all interior components requiring service, maintenance, or adjustment, shall in-no way interfere with the launching ofa target or the use of the me- . chanical and electrical safety -pins.:

3.3.4 Pressurization System. The pressurization system shall have a capacity of 150 cubic inches and shall be capable of being charged to a pressure of at least 3000 pounds per square inch (psi) at a temperature of 70° Fahrenheit (F) with nitrogen conforming to BB-N-411, Type I, Grade A. The general arrangement shall be as shown in Figure 2.

3.3..4.1 Receiver Assembly. The nitrogen receiver assembly shall incorporate an indicator to identify a pressurized system and. indicate the amount of pressurization; a device which will prevent jet propulsion of the receiver should the control assembly be broken when the receiver assembly is subjected to a forty foot drop test; and a fire burnout and pressure burst disc that willbe the first component to fail when the receiver assembly is subjectedto abnormal heat due to proximity to fire: temperature failure shall be between 185° and 215° l?, pressure release shall be between 4400 and 46oo psi.

3.3.461.1 Receiver (Bottle). The bottle (reservoir) of the receiver assembly shall be in accordance with MIL-R-8573 Type 30-45, Class A, Designation 150C.

3.3.4.1.2 Regulator Valve Assembly. The regulator valve assembly shall supply a constant gas flow, regardless of altitude, of 0.0205 pounds per minute (ppm) for not less than 60 minutes. The regulator valve operating range shall be factory adjusted between 3.76 pounds per square inch gage (psig) and 5.73 psig and the outlet orifice shall be factory set to the 0.0205 ppm flow under standard National Advisory Committee for Aeronautics conditions at the barometric pressure equivalent of 50,000 feet (ft) altitude.

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3.3.4.2 Barometric Switch. The pressurization system shall be controlled by a barometric switch which automatically actuates the system at altitudes in excess of 10,000 ft, allowing flow of nitrogen to the target. The switch shall automatically deactivate the system when the aircraft descends below an altitude of 9ooo ft.

3.3.5 Control Box. Electrical power and operating signals are received from the aircraft through a connector. The control box shall provide for the distribution of the power and signals to the launcher for its operation and to the target for its pre-launch requirements. Three relays (Select Power, Pitot Heater, and Launch Cartridge) shall be provided in the box. The box shall also provide two resistors, one test point, and the electrical safety switch. When tested in accordance with 4.4.22, the electrical resistance of closed circuits shall be not less than 0.5 ohm. The resistance of open circuits shall be infinity.

3.3.6 Cartridge, The launcher operation shall be actuatedby the ignition of a Mk 9 Mod 0 cartridge conforming to MIL-C-23931.

3.3.7 Input Power. The launcher shall be capable of operating satisfactorily when supplied, by the aircraft, with direct current (d.c.) electric power having the characteristics specified in MIL-STD-704 as Category "A".

3.3.8 Safety Devices. Two safety devices shall be incorporated in the launcher to prevent launching of the target on the ground. A mechanical safety pin shall lock the retaining hooks and an electrical safety pin shall ground the cartridge firing circuit.

3,3.8.1 Venting. The launcher shall be provided with means to vent the products of combustion of cartridge actuated devices in a manner that will. preclude the ignition of an ambient explosive atmosphere.

3*3*9 Check-Out. The launcher shall be compatible with and capable of being checked out by the TTu-183/F test set described in NAVWEPS 17-15-26.

3.3.10 Operability. When tested in accordance with 4.4.25, the launcher shall not stick, bind, or display other evidence of malfunction and shall respond as defined in 6.2.2.

3.4 Performance Requirements.

3.4.1 Target Launch. The launcher shall be capable of launching the target as the aircraft covers the range from a minimum altitude of 1000 ft and speed of Mach 0.6 to a maximum altitude of 50,000 ft and speed of Mach 2,0.

3.4.2 Launch Acceleration. Target vertical launch acceleration shall not exceed **12 g.**

3.4.3 Separation Velocity. Target vertical separation velocity shall be not less than *18* feet per second (fps).

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3*5 Environmental Requirements.

3.5.1 Temperature, Storage. When temperature tested in accordance with 4.4.2, the launcher shall show no evidence of damage.

3.5.2 Temperature, Altitude. When temperature and altitude tested in accordance with 4. 4.3, the launcher shall show no evidence of damage and shall operate satisfactorily.

3.5.3 Vibration. When vibration tested in accordance with 4.4.4, the launcher shall. show no evidence of damage and shall operate satisfactorily.

3.5.4 Shock. When shock tested in accordance with 4.4.5, the launcher shall show no evidence of damage and shall operate satisfactorily after each shock has been imposed.

3.5.5 Salt Spray. When tested in accordance with 4.4.6, the launcher shall show no evidence of corrosion or damage and shall operate satisfactorily upon completion of the test.

3.5.6 Sand and Dust. When tested in accordance with 4.4.7, the launcher shall show no evidence of damage and shall operate satisfactorily upon completion of the test,

3.597 Radio Interference. When tested for the effects of radiated or conducted radio interference in accordance with 4.4.8, the launcher shall respond within the limits imposed by MIL-I-6181. .

3.5.8 Static Strength. When tested in accordance with 4.4.9, the launcher shall not release the dummy target except when operated. After the limit and yield loads have been applied and released, there shall be no structural damage. Application and release of limit loads shall result in no deformation,

3.5*9 Icing . When tested for operation under icing conditions in accordnce with 4.4 .10, the launcher shall operate satisfactorily.

3.5.10 High g release. When tested for high g release in accordance with 4.4.11, the launcher shall operate satisfactorily.

3.5.11 Lag or Time of Release Determination. When tested to determine the lag or time of release in accordance with 4. 4.12, the launcher shall fall within the limits imposed by MIL-T-7743.

3.5.12 Operating Life. When tested for operating life in accordance with 4.4.13, the launcher shall display the reliability characteristics required by 3.2.14.

3.6 Shipping Condition.

3.6.1 Shock, Transportation, and Storage. After being subjected to the"

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3.6.1 (Cont'd)

test of 4.4.26.1, the launcher shall show no damage and shall. operate satisfactorily.

3.6.2 Vibration, Transportation. After being subjected to the test of 4.4.26.2, the launcher shall show no damage and shall operate satisfactorily.

3*7 Cleanliness. Prior to and after final assembly, the launcher shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, rosin flux, and other foreign material. Burrs and sharp edges shall be removed.

3.8 Workmanship. The launcher, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, tiring, and marking of parts and assemblies, and to plating, machine screw assembly, welding, and brazing, where applicable. The standards of workmanship exhibited in any approved preproduction sample shall be determinative of the requirements of the contract relative to workmanship.

4 QUALITIY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections *are* deemed necessary to **assure** supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests.Equipments shall be subjected to the following tests to determine compliance with all the applicable requirements:

a. Preproduction Tests (4.2)
b. Acceptance Tests (4.3)

4.1.2 Test Samples. The launchers used for preproduction and sampling tests shall. not be considered as equipments under the contract. They are to be supplied as equipments over and above-the quantity specified for delivery in the contract. Samples may be returned to the contractor or retained at the option of the procuring agency.

4.1.3 Test Conditions. Unless otherwise specified, all tests shall be performed at the following conditions:

a. Room Temperature - Room ambient of 75° ± 10°F

b. Test Temperature - All parts shall be completely stabilized at the specified temperature prior to conducting any tests.

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4.2 Preproduction Tests. Preproduction tests are those tests which are conducted by the contractor and the Government on the first three launchers produced from production methods and tools. These preproduction samples shall have successfully passed the individual tests of 4.3.1 conducted by the contractor prior to their delivery, at the contractor's expense, to a Government selected test laboratory. When the contractor delivers the sample units to a Government selected test laboratory, a copy of the individual tests results together with a certification from the cognizant Government inspector that the materials used in the samples are in accordance with the requirements of this specification shall be forwarded with the samples. Any defects or failures occurring during the Government tests shall be cause for tentative rejection of the preproduction samples. All rejected samples may be returned to the contractor for rework or correction of defects when considered practical and feasible by the procuring agency, otherwise the contractor will be required to submit additional samples for retest in lieu of the reworked samples. Preproduction tests shall consist of all acceptance tests plus the tests listed in Table I.

TABLE I

Sample Number	1	2	3
Launch acceleration & separation velocity			X
Temperature, storage	X		
Temperature, altitude	X		
Vibration	X		
Shock	X		
Salt spray		X	
Sand and dust		X	
Radio interference			X
Static strength			X
Icing		X	
High g release		X	
Lag or time of release		x	
Operating Life		x	
Safety devices		X	

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4.2.1 Preproduction Approval. Approval of the preproduction samples shall be by the procuring agency upon satisfactory completion of all tests. Any subsequent changes in design or configuration shall be in accordance with ANA 445 and, Supplement No, 1 thereto, No production equipments or lot samples shall be delivered prior to approval of the preproduction samples. Prefabrication of product equipment prior to the approval of the preproduction models is at the contractor's own risk, The approved preproduction samples may be returnedto the contractor for his own use in the fabrication and. testing of equipment to be submitted for acceptance.

4.3 Acceptance Tests. Acceptance of equipment shall be on a lot basis. Each lot shall be composed of equipment which has successfully passed the individual tests. Acceptance tests shall consist of the following:

a, Individual Tests
b, Sampling Tests

4.3.1 Individual Tests. Individual tests shall be accomplished, under the responsibility of the contractor, on each equipment submitted for acceptance under the contract. The contractor shall maintain test reports showing the quantitative results for all. individual tests, Such reports shall be signed by an authorized representative of the contractor and this data shall be made available to the procuring agency upon request. Individual tests shall consist of the following tests:

Control box
a. Launcher check-out
c. Electrical safety device
d. Operability

4,3.2 Sampling Tests. Sampling tests shall be conducted at a Government designated laboratory facility under the responsibility of the procuring agency. Sampling tests shall be conducted. on samples selected. from lots of equipment which have successfully passed the individual tests, Samples to be tested shall be selectedby the Government Inspector in accordance with the following sampling test schedule (4.3.2.1) and. forwarded. to the Government selected laboratory for testing. A lot shall not exceed" 300 launchers except with the approval of the procuring agency. Samples shall be suitably labeled to identify therewith a particular lot, Sampling tests shall consist of the following examinations and tests:

Examinations

	Requirements
a. Materials	3.2.1
b. Fungus resistance	3.2.2
Protective coatings	3.2,3
d. Marking	3.2.4
e. Lubrication	3.2,5
f. Threads	3.2,6
Electrical connectors	3.2.7
h: Relays	3.2,8

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Examinations	Requirements
i. Wiring	3.2.9
j. Heat treatment	3,2.12
k. Special tools	3.2.15
1. Access doors	3.3.3
m. Venting	3.3.8.1
n. Cleanliness	3.7
o. Workmanship	3.8

Tests	Requirements
a. Dielectric strength	3.2.10
b. Insulation resistance	3.2.11
c. Interchangeability	3,2.13
d. Dimensions	3.3.1
e. Weight	3.3.2
f. Pressurization system	3,3.4
g. Barometric switch	3.3.4.2
h. Control box	3*3*5
i. Electrical safety device	3.3.8
Launcher check-out	3*3*9
k. Operability	3.3.10
l. Shipping condition	3.6

4.3.2.1 Sampling Test Schedule. Unless otherwise specified in the contract, the sampling. test schedule shall be as follows :

Quantity offered for lot acceptance	Quantity to be tested
99 or less	2
100 to 299	3
300 to 499	4
500 or more	4 plus 1 for each additional 500 or fraction thereof

4.3.2.2 Limitation of Lot Size, Lot sizes shall. be as specified in the contract. Lots shall be limited to launchers having the same nomenclature and part number produced by the same manufacturer using the same parts, materials, processes, and production facilities in continuous production.

4.3.3 Approval of Production Lots. Approval of production lots shall be by the procuring agency upon satisfactory completion of all sampling tests. See 4.5 and 4.6.

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4.4 Test Procedures,

4.4.1 Launch Acceleration and Separation Velocity. With the dummy target attached, the launcher shall be operated. Determine the target vertical acceleration and the target vertical separation velocity.

4.4.2 Temperature, Storage.

a. The internal temperature of the test chamber shall be reduced to minus 80° F and the launcher inserted therein, The temperature of the launcher shall be stabilized at minus 80° F and shall then be maintained at that temperature for 72 hours. At the end of this period the launcher shall be removed from the chamber and visually examined for physical damage.

b. The internal temperature of the test chamber shall then be raised to plus 160° F and the launcher re-inserted therein. The temperature of the launcher shall be stabilized at plus 160° F and shall then be maintained at that temperature for 72 hours . At the end of this period the launcher shall be visually examined for physical damage.

4.4.3 Temperature, Altitude.

a. The launcher with dummy target (see Figure 3) attached (see 6.2.1) shall be inserted in the test chamber. The temperature and pressure shall be stabilized at ambient conditions.

b. The launcher shall be operated (see 6.2.2) and observed for evidence of sticking or binding, The launcher shall be removed from the test chamber and visually examined for physical damage.

c. The internal temperature of the test chamber shall be reduced to minus 65° F with the pressure at normal ground level. The launcher shall be returned to the chamber and the dummy target shall be re-attached. Their temperature shall be stabilized. Repeat step b,

d. The internal temperature of the chamber shall be maintained at minus 65° F. The launcher shall be returned to the chamber and the dummy target shall be re-attached. With the temperature stabilized, the chamber pressure shall be reduced to the equivalent of a 60,000 ft altitude. Repeat procedure b.

e. Repeat step d. except that the temperature shall be minus 65° F and the pressure shall be the equivalent of a 35,000 ft altitude.

f. Repeat step d. except that the temperature shall be plus 160° F and the pressure shall be the equivalent of a 35,000 ft altitude.

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4.4.3 (Cent'd)

g" Repeat step d. except that the temperature shall be plus 160° F and the pressure shall be the equivalent of a 60,000 ft altitude.

4.4.4 Vibration. The launcher shall be tested in accordance with the vibration test of MIL-T-7743, except that the maximum frequency of the vibration shall be 500 cycles per second (cps) during the resonance vibration test. The support points shall be the attachment points between the dummy target and the launcher.

4.4.5 Shock. The launcher shall be subjected to the shock tests ofMlL-T-7743 . Cartridges shall be in the breech during the tests. The launcher shall be operated immediately after each test.

4.4.6 Salt Spray. The launcher shall be subjected to the salt spray test of MIL-E-5272, except that the launcher shall not be re-operated and inspected until 48 hours after the completion of the test. A 5% salt solution shall be used. Cartridges shall be inserted in the breech of the launcher immediately prior to operation of the launcher. The launcher shall be operated immediately after the test.

4.4.7 Sand and Dust. The launcher shall be subjected to the sand test of MIL-T-7743, except that the duration shall be 3 hours, the particle velocity shall be 2,500 ± 500 feet per minute, and no frozen sand test shall be performed. Cartridges shall be inserted in the breech of the launcher immediately prior to operation of the launcher. The launcher shall be operated immediately after exposure to the sand test.

4.4.8 Radio Interference. Conducted or radiated radio interference generated by the launcher as well as susceptibility of the launcher to conducted or radiated radio interference shall be tested in accordance with MIL-I-6181.

4.4.9 Static Strength. The launcher shall be subjected to the loads listed in Figure 4. The loads shall be applied to the dummy target and the launcher as indicated in Figure 5 and the following test procedure:

a, Attach the dummy target to the launcher.

b. Apply loads A-1 andA-2; then apply loads A-3 through A-9.

c. Increase loads A-1 andA-2 to B-1 andB-2, respectively; then increase A-3 through A-9 to B-3 through B-9, respectively,

d. Increase loads B-1 and B-2 to C-1 and C-2; then increase B-3 through B-9to C-3 through C-9.

e. Release all loads.

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4.4.9 (Cent'd)

f. Measure and record to the nearest 1/8 inch the deflection in the X, Y, and Z directions of points M-1 through M-6 shown on Figure 5.

g. Operate the launcher.

h, Inspect the launcher for physical damage.

i. Re-attach the dummy target.

j Apply loads D-1 and D-2; then apply loads D-3 through D-9.

k, Repeat steps e., g., and h. Re-attach the dummy target,

1. Apply loads E-1 and E-2; then apply loads E-3 through E-9.

m. Increase loads E-1 and E-2 toF-1 and F-2; then increase E-3 through E-9 to F-3 through F-9.

n, Increase loads F-1 andF-2 to G-1 and G-2; then increase F-3 through F-9 to G-3 through G-9. Maintain loads for a minimum of one minute.

0. Release all loads.

P" De-torque the sway brace bolts and release the dummy target by manually operating the launcher.

4.4.10 Icing . The launcher shall be tested in accordance with the icing test of MIL-T-7743. The cartridges shallbe inserted in the breech immediately prior to launcher operation. The launcher shall be operated immediately after the completion of the test.

4.4.11 High g release. The launcher shall be tested to determine that it operates satisfactorily under the high g release test of MIL-T-7743 except that the number of releases shall be 5 instead of 25 and (b) shall be deleted.

4.4.12 Lag or Time of Release Determination. The launcher shall be tested in accordance with MIL-T-7743, except that only the ejection system tests and horizontal release tests shall be performed.

4.4.13 Operating Life. The launcher shall be tested as follows (observe launcher for proper operation):

a. 10 operations without bench check or cleaning

b. 100 operations (including a.) with a bench check and cleaning after each 10 operations

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4.4.13 (cont'd)

c. 500 operations (including.) with an overhaul at the end of each 100 operations

4.4.14 Safety Devices. With the dummy target attached, the cartridge inserted in the breech, and the electrical and mechanical safety pins in place,-the launcher shall be supplied with normal operating power. The cartridge firing circuit shall be closed. Observe cartridge for no ignition. Open the cartridge firing circuit. Remove the electrical safety pin. Close cartridge firing circuit. Observe launcher for restraint of dummy target.

NOTE ; This test shallbe performed as the final operation of 4.4.13.

4.4.15 Dielectric Strength. An electromotive force of at least 1056 volts root mean square (rms), single phase alternating current (a.c.), 50-100 cps shall be applied for one minute to the electrical insulation between each electrical circuit and all other circuits and between each electrical circuit and the structural ground. An a.c. voltage not to exceed 500 volts rms shall be applied between relay contacts. Observe for leakage, arcing, breaks, or other damage. The cartridge shall not be inserted.

4.4.16 Insulation Resistance. A constant voltage of 500 volts d.c. shall be applied between each electrical circuit and device and all other electrical circuits and devices and between each circuit and device and the structural ground. " Measure the insulation resistance of the circuits and devices.

4.4.17 Interchangeability, From each sample consisting of two or more launchers, parts, components, and subassemblies shall be chosen at random and assembled. Observe for mis-mating or non-fit.

4.4.18 Dimensions. The launcher shall be measured to show compliance with the dimensions and tolerances shown on NAVAIR Dwg. 67A75H1.

4.4.19 Weight. Weigh the fully assembled launcher in accordance with industry practices.

4.4.2o Pressurization System. The pressurization system shall be charged to 3000 psi with nitrogen and then activated. The delivery pressure in psig and the flow rate in ppm shall be measured at one minute intervals for 60 minutes.

4.4.21 Barometric Switch: The assembled launcher shall be placed in a test chamber and electrical power applied. The pressure within the chamber shall be reduced to the equivalent of a 9000 ft altitude. The pressure shall thenbe further reduced to the equivalent of' a 10,000 ft altitude. The pressure shall then be alternated between 9000 ft and 10,000 ft equivalents. Note the pressures at which the barometric switch operates.

4.4.22 Control Box. Prior to assembly of the control box into the launcher, the electrical circuits in the box shall be checked for continuity, the electrical resistance of the circuits shall be measured, and the operation of all relays shall be tested.

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4.4.23 Electrical Safety Device. With the dummy target attached, the cartridge inserted in the breech, and the electrical safety pin in place, the launcher shall be supplied with normal operating power. The cartridge firing circuit shall be closed. Observe cartridge for no ignition.

4.4.24 Check-Out. Mate the TTu-183/F test set with the launcher. Observe *for ease* and rapidity of attachment. With normal input power supplied, use the test set to check out the launcher for readiness for operation.

4.4.25 Operability. With the dummy target attached, the launcher shallbe actuated with nitrogen supplied to the cartridge chamber as a power source. Mechanical components shall be observed for evidence of sticking or binding. Reattach the dummy target to the launcher and operate the launcher. Observe for cartridge ignition and launcher operation.

4.4.26 Shipping Condition.

4.4.26.1 Shock, Transportation, and Storage. The launcher, packaged and packed as specified in Section 5, shall be subjected to the appropriate shock isolation test specified in WR-11.

4.4.26.2 Vibration, Transportation. The launcher, packaged and packed as specified in Section 5, shall be vibration tested as specified in WR-11.

4.4.26.3 Functional Tests. At the conclusion of the tests specified in 4.4.26.1 and 4.4.26.2, tests 4.4.20, 4.4.21, 4.4.22, 4.4.23, and 4.4.25 shallbe repeated.

4.4.26.4 Waiver of Tests. Tests specified in 4.4.26.1 and 4.4.26.2 maybe waived by the procuring activity for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.5 Rejections.

4.5.1 Production Lot Sample Rejection. The failure of any production lot sample to pass the inspection and tests outlined in this specification shall cause tentative rejection of all launchers representedby that sample, subject to *retest as* hereinafter provided.

4.5.2 Individual Production Unit Rejection. The failure of any launcher to pass the individual inspection and tests outlined in 4.3.1 shall cause its tentative rejection, subject to retest as hereinafter provided.

4.5.3 Salvaged Pieces. Any piece(s) from a rejected launcher may be **salvaged** for use providing each salvaged piece is inspected and is foundto be in complete compliance with the requirements of this specification and the associated drawings and specifications.

4 . 6 Retests and Re-inspection.

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4.6.1 Production Lot Samples - First Retest. In the event any sample representing a production lot fails to pass the inspection and tests outlined herein, the contractor shall have the opportunity of reworking the entire affected lot and resubmitting to retest twice as many samples, selected at random by the Naval inspector, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be rejected.

4.6.2 Production Lot Samples - Second Retest. If, in the contractor's opinion, the failure of the retest samples is due to a remedial defect. the contractor may, at the discretion of the Naval inspector, be offered the opportunity of reworking the entire affected lot and resubmitting to a second retest three times as many samples, selected at random, as originally submitted for the lot. Should all of these samples pass the required tests, the lot represented thereby shall be considered acceptable. Should any of these samples fail, the entire lot represented thereby shall be finally rejected.

4.6.3 Retest and Re-Inspection of Individual Units, Any launcher which fails to pass the individual tests as outlined in 4.3.1 may, upon the request of the contractor, be resubmitted for individual inspection and test, provided a statement showing the corrective measures taken is furnished to the Inspector at the time of such resubmission. Upon re-inspection and retest, failure to meet the requirements outlined herein shall be cause for rejection of the unit.

4.6.4 Inspection and Testing Subsequent to Delivery. Any or all launchers may be subject to inspection and testing for conformance with any or all requirements and tests of this specification after delivery to the Government.

5 PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. When furnished by the Government, a reusable shipping container shall be used for packaging and packing the launcher in lieu of the requirements of paragraphs 5,1.1.4 through 5.2.2. Should a container not be furnished by the Government, paragraphs 5.1.1.4 through 5.2.2 shall apply.

5.1.l Level A.

5.1.1.1 The launcher shall be cleaned in accordance with Method c-1 of MIL-P-116 .

5.1.1.2 Drying. "The launcher shall be dried in accordance with Method D-4 of MIL-P-116.

5.1.1.3 Preservative Application. Exposed metal surfaces subject to corrosion and not otherwise treated shall be coated with one application of Preservative P-2 of MIL-P-116 applied by brushing or spraying.

5.1.1.4 Unit Packaging. The unit packaging of the launcher shall be in accordance with Method IA-13 of MIL-P-116. When wood or wood-cleated containers are used as the outer container, they shall be of the overseas type and shall be lined with fiberboard in accordance with PPP-B-636, Type I, Class 2.

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5,1.1.5 Cushioning. At the time of packaging or packing, the moisture content of the cushioning material shall be not greater than 5% and the acid level (hydrogen ion content, pH) range of the material shall be between 6.0 and 8.0.

5.1.2 Level C.

5.1.2.1 Cleaning. As in 5.1.l.1,

5.1.2.2 Drying As in 5.1.1.2.

5.1.2,3 Preservative Application. As in5.1.l.3.

5.1.2.4 Unit Packaging. Unit packaging of the launcher shall be in accordance with Method III of MIL-P-116, The unit package shall contain one launcher.

5.2 Packing.

5.2.1 Level A. Launchers packaged as described in 5.1.1.4 shall. require no overpacking when wood or wood-cleated boxes are used as the outer container for Method IA-13 packages, When the outer container of the package is a fiber-board box, the package shall. be packed in wood boxes in accordance with PPP-B-601, overseas type.

5.2.2 Level C. Launchers packaged in accordance with 5,1.2.4 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use, Containers shall comply with the Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5*3 Marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, and shall include serial number, date of manufacture, and "Property of U. S. Government".

6 Noms

6.1 Supplementa ry Information. Information pertaining to the AQM-37A Missile Target may be obtained from the Naval Air Systems Command (AIR-5351) , Department of the Navy, Washington, D. C. 20360.

6.2 Definitions.

6.2.1 Dummy Target Attachment. The attachment of a dummy target to a launcher includes the torquing of the 4 launcher sway brace adjustment bolts to 30 ± 5 inch-pounds and the tightening of the related lock nuts.

6.2.2 Launcher Operation. Launcher operation is defined as the cartridge-actuated catapulting of the dummy target from the launcher when it is supplied with the input power described in 3.3.7, followed by the return of the sled and its retention in the retracted position.

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6.3 Ordering Data. Procurement documents for the launchers should specify the following:

a. Title, number, and date of this specification

b. The number of launchers to be produced and the schedule of delivery

c. Technical data requirements:

(1) Drawings, ID's (DL's), etc.
(2) Specification (subordinate)
(3) Microfilm
(4) Factory Acceptance Test Plans& Procedures
(5) preproduction Test Plans &Procedures
(6) Ordnance Publications, Manuals, etc. required
(7) Serial number assignment

d. Options (Attention is invited to the Quality Assurance Provisions and Options in MIL-P-116 and to 4.4.26.4.)

e. Change approval procedures

f. Precedence of documents

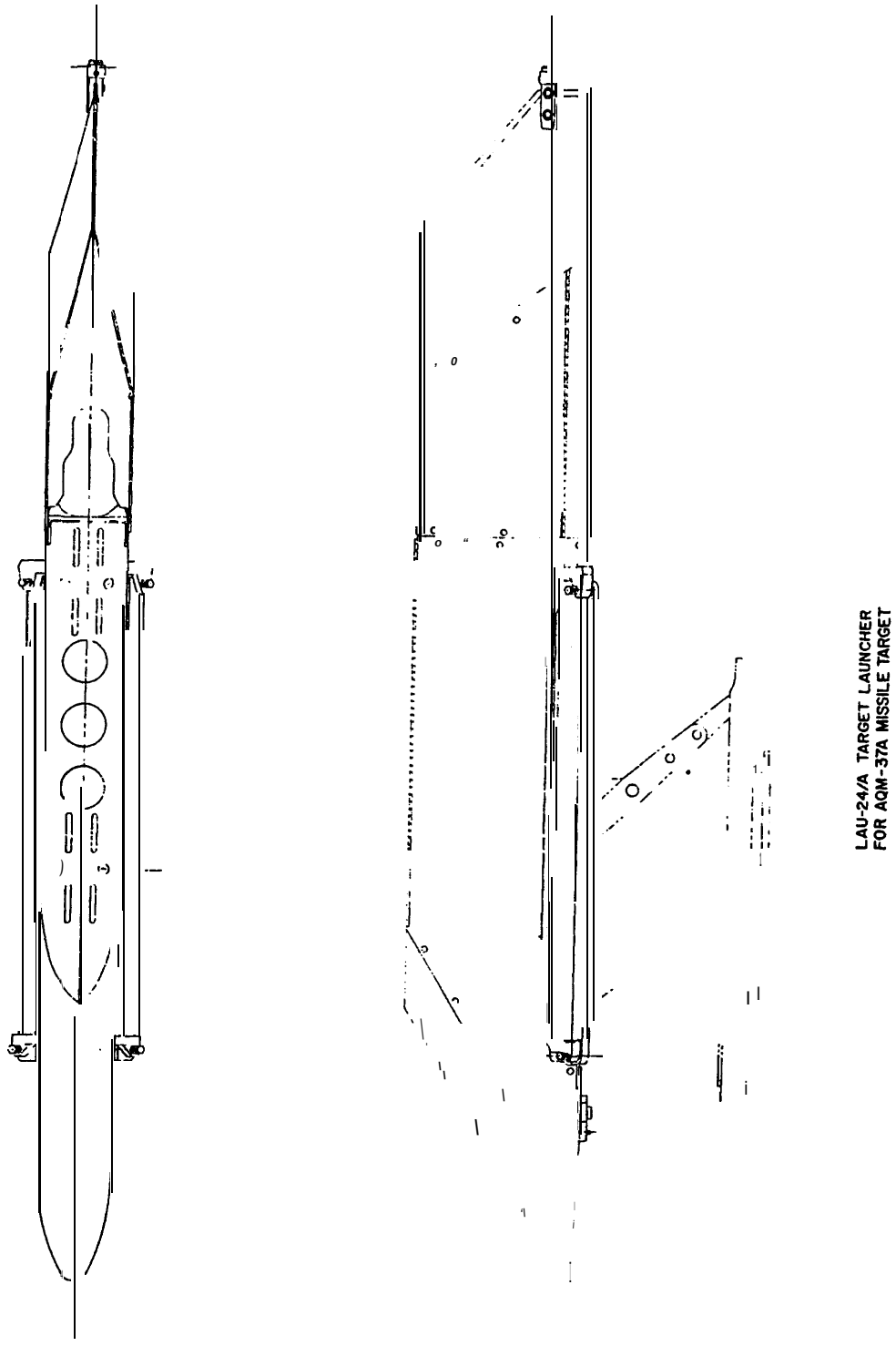
g. Levels required in preparation for delivery. Criteria for choice of level are:

Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for re-distribution anywhere.

Level-C. This level shall. be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.4 Superior Substitutes. Recommendations concerning the use of a superior substitute for any part or material specified herein shall be accompanied by a description of the proposed substitute, the reasons therefor, together with . evidence to substantiate any claim as to its suitability, and shall be submitted to NASC for approval. At the discretion of NASC, test samples may be required to prove the suitability of the proposed substitute-.

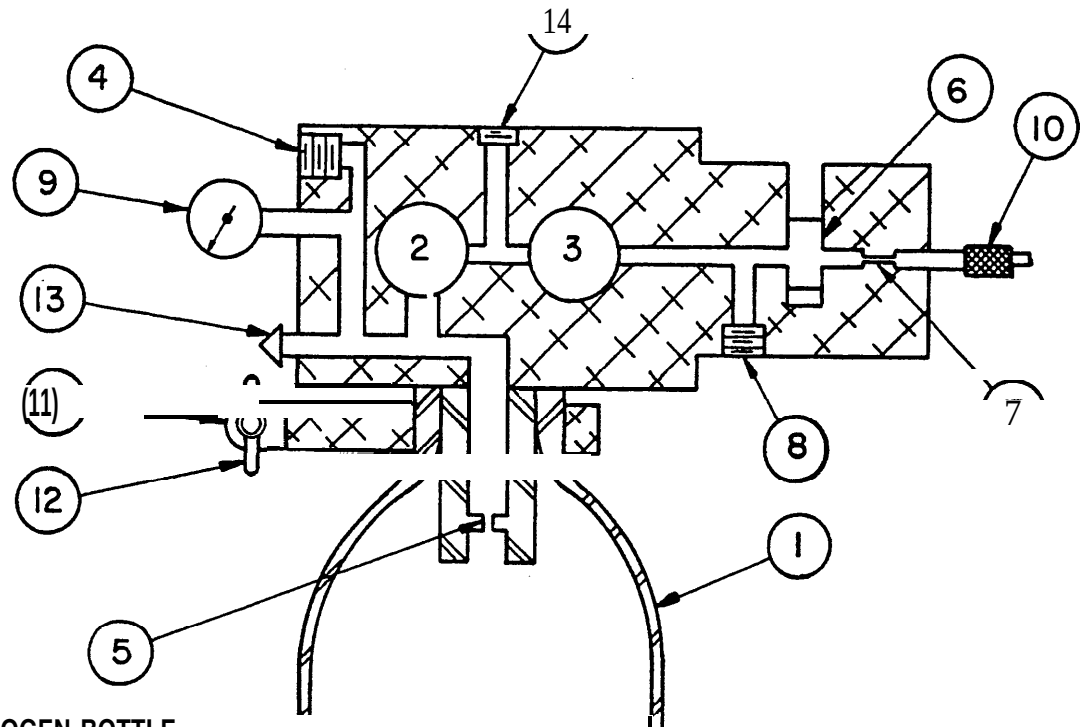
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LAU-24/A TARGET LAUNCHER
FOR AQM-37A MISSILE TARGET

FIGURE I

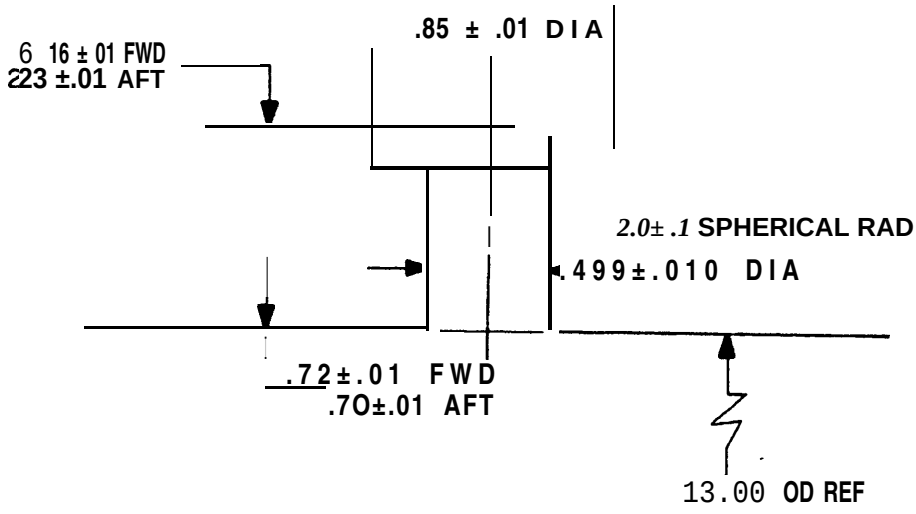
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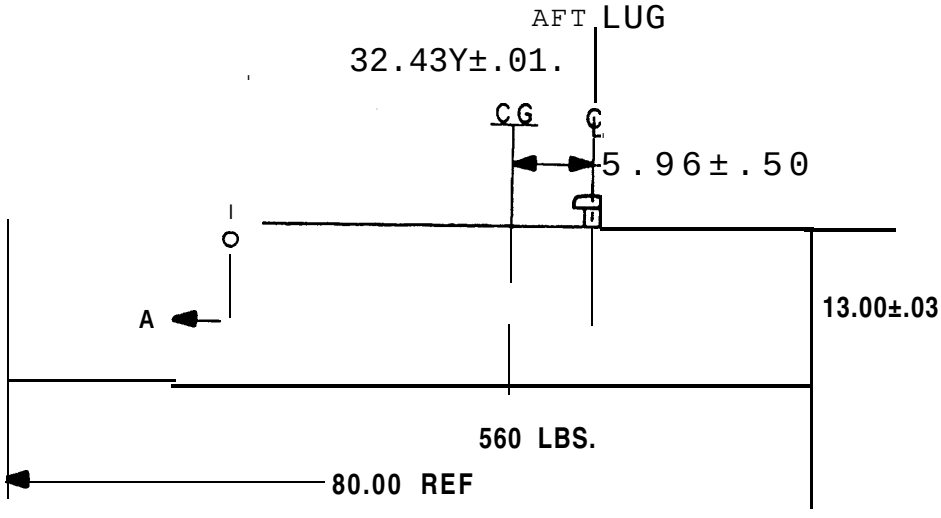
1. NITROGEN BOTTLE
2. HIGH PRESSURE REGULATOR VALVE
3. LOW PRESSURE REGULATOR VALVE
4. FIRE BURNOUT AND PRESSURE BURST DISC
5. FLOW ORIFICE
6. ELECTRICALLY ACTIVATED LOW PRESSURE SHUTOFF VALVE
7. LOW PRESSURE ORIFICE
8. LOW PRESSURE RELIEF VALVE
9. DIAL INDICATOR
10. QUICK RELEASE CONNECTOR
11. MOUNTING CLAMP
12. QUICK RELEASE MOUNTING PIN
13. NITROGEN BOTTLE CHARGING VALVE MS 28889-1
14. INTERMEDIATE RELIEF VALVE

NITROGEN RECEIVER ASSEMBLY
FIGURE 2

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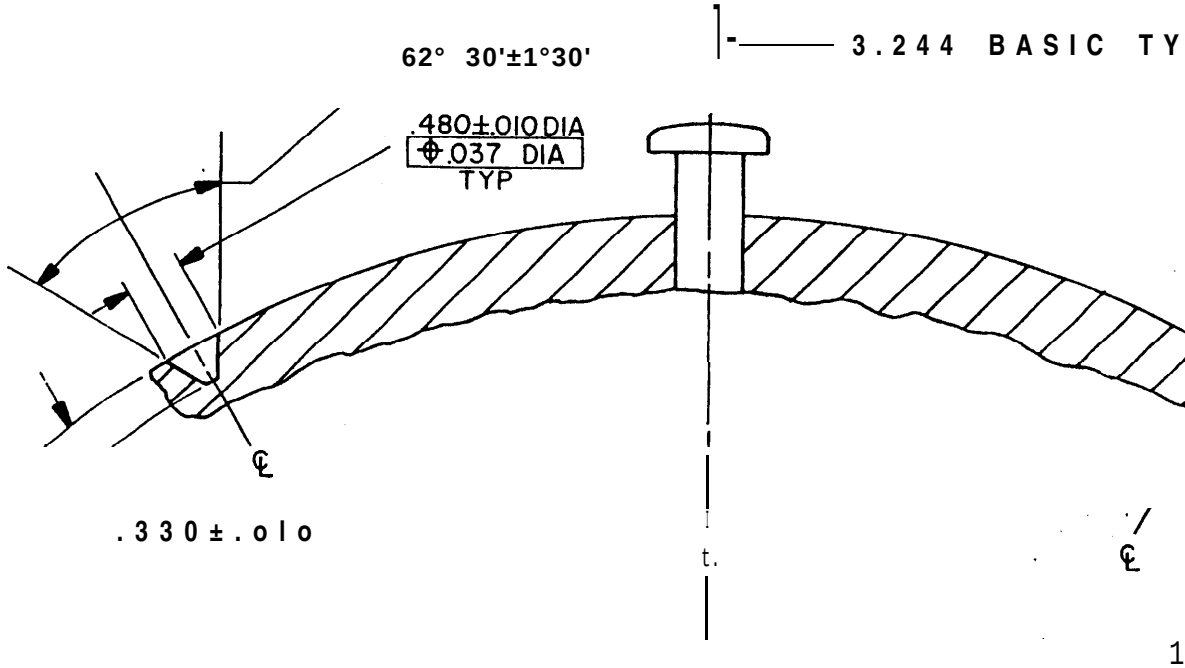
FORWARD AND AFT MOUNTING LUGS "
AFT LUG HAS DASHED(---)PORTION REMOVED



DUMMY TARGET
FIGURE 3, SHEET I

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FIGURE 0



SECTION A-A

DUMMY TARGET

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% ULT. LOAD	TYPE OF LOAD	IDENT. No.	LOAD IN POUNDS							
			VZPI.	VZPR	VXP	vyP	VyA IrPF Fwd	VyAirPA Aft	Vzs	Vxs
			1	2	3	4	5	6	7	8
10%		A	+18.95	+18.95	+42.00	+136.80	+lo2.45	+37.75	+302.00	+206.
50%.		B	94.75	94.75	210.00	684.00	512.27	188.73	1510.00	1030
66. 67%	LIMIT	C	126.34	126.34	280.00	912.04	683.06	251.65	2013.43	1373
76. 67%	YIELD	D	145.29	145.29	322.00	1048.84	785.51	289.40	2315.43	1579
87.500/.		E	165.81	165. 81	367.50	1197.00	896.47	330.28	2642.50	1802
93.757*		F	177.66	177.66	393.75	1282.50	960.50	353.87	2831.25	1931
.00.00%	ULT.	G	189.50	189.50	420.00	1368.00	1024.54	377.46	3020.00	2060

FIGURE 4

STATIC STRENGTH TEST' LOADS

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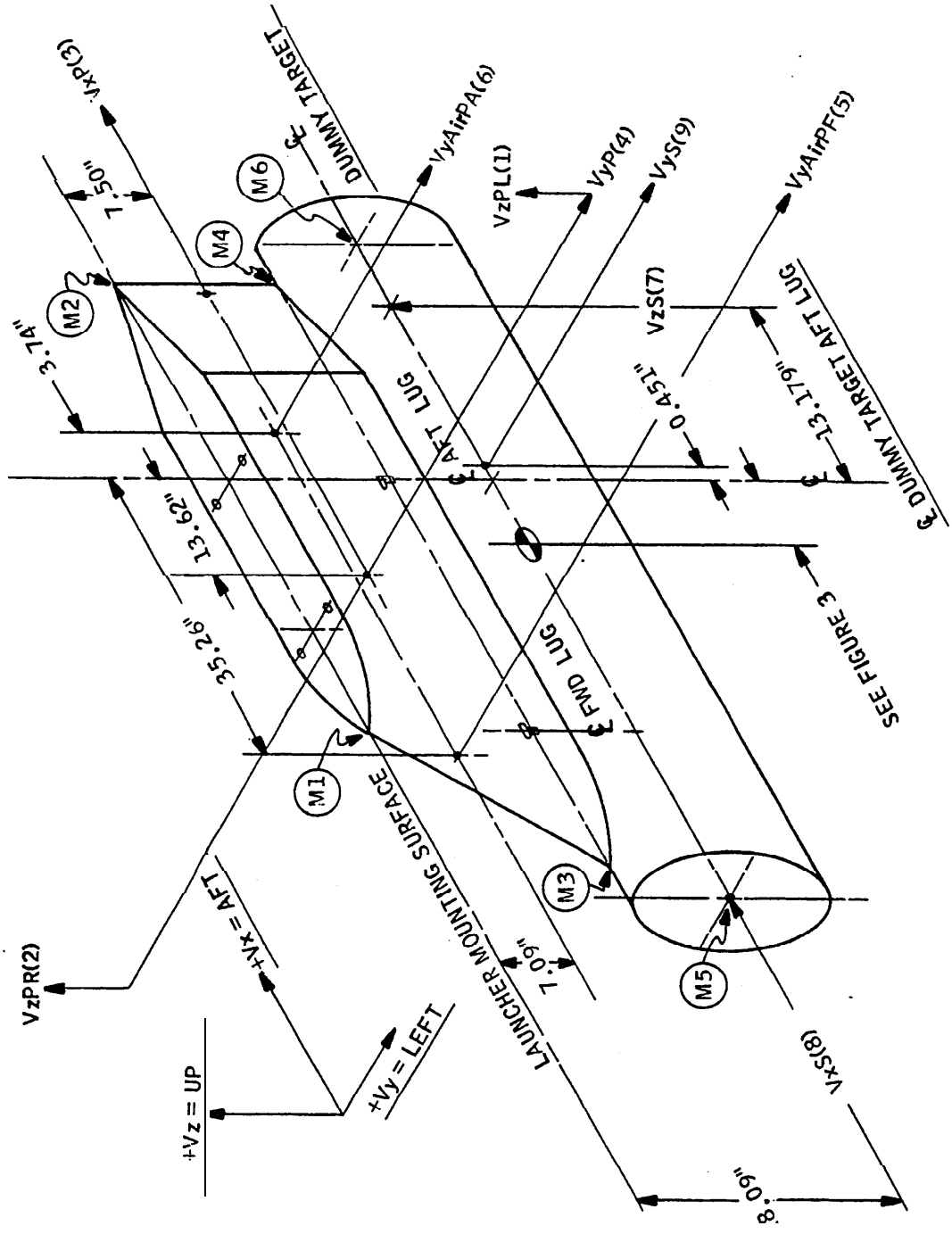


FIGURE 5

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SPECIFICATION ANALYSIS SHEET

FOrm Approved
Budget Bureau No. 119-Ro04

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of th ification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-L-81520(AS) LAUNCHER, TARGET, LAU-24/A FOR USE WITH AQM-37A MISSILE TARGET

ORGANIZATION (of submittcr)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

❑ DIRECT GOVERNMENT CONTRACT o SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REWIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B, RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVEI

n YES n No IF 'YES", IN WHAT WAY?

REMARKS *(Attach any* pertinentdatawhich maybe ot.use in..improving this specification. *If* there are additional *papers, attach to form andplace both* in an *envelope addressed* to *preparing* activity)

SUBMITTED BY *(Prtnted or-typed name and* activity)	DATE

DD, FORM 1 APR 63 1426

C-8878

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FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20360

PO STAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND (AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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